

$$R_{n+1}(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi) (x-a)^{n+1}$$

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$$f(x) = e^x, a=2, [a, \beta] = [0, 3] \tau = 1/10$$

$$R_{n+1}(x) = \frac{1}{(n+1)!} e^{\xi} (x-2)^{n+1} \rightarrow 0 < x < 3 \rightarrow -2 < x-2 < 1 \rightarrow -2 < x-2 < 2$$

$$|x-2| < 2$$

$$0 < x < 3 \rightarrow -3 < x-2 < 3 \rightarrow e^{-3} < e^{\xi} < e^3$$

$$|e^3| < e^3$$

$$\text{plug } n: \frac{1}{10} = \frac{1}{(n+1)!} e^3 2^{n+1}$$

$$\frac{(n+1)!}{2^{n+1}} = 10 e^3 \approx 200.95$$

$$\text{for } n=5: \frac{720}{64} \approx 11.25$$

$$\rightarrow (n=9) \text{ see left}$$

n=0	2	1
n=1	4	2
n=2	8	6
n=3	16	24
n=4	32	120
n=5	64	720
n=6	128	5040
n=7	256	40320
n=8	512	362880
n=9	1024	

$$\frac{40320}{256}$$

$$f(x) = \cos(2x), a=0, [a, \beta] = [-\frac{\pi}{4}, \frac{\pi}{4}] \tau = 1/10$$

$$f'(x) = -2\sin(2x)$$

$$f''(x) = -4\cos(2x)$$

$$f'''(x) = 8\sin(2x)$$

$$f^{(4)}(x) = 16\cos(2x)$$

$$f^{(n+1)}(x) = 2^{n+1} \cos(2x + (n+1)(\frac{\pi}{2}))$$

$$\rightarrow R_{n+1}(x) = \frac{f^{(n+1)}(\xi) (x-a)^{n+1}}{(n+1)!} = \frac{2^{n+1} \cos(2\xi + (n+1)(\frac{\pi}{2})) (x-a)^{n+1}}{(n+1)!}$$

max of cos is 1 on the interval

$$\text{max of } x \text{ on interval is } \frac{\pi}{4} \rightarrow \text{plug } n: \frac{2^{n+1} (\frac{\pi}{4})^{n+1}}{(n+1)!} = \frac{1}{10} \rightarrow (n=4)$$

$$\cos(x + \frac{\pi}{2}) = -\sin(x)$$